Single-Image Pixel Fraction Report

Single-image AlphaEarth-to-pixel-value feasibility experiment.

Input CSV: DataSources/single_image_pixel_fraction/sentinel2_alphaearth_pixel_pairs_2024_n5000

Model: StandardScaler + Ridge(alpha=1.0)

Target band: B4

Train rows: 3635

Test rows: 909

Best result on this split:

Fraction = 0.50

R^2 = 0.676

RMSE = 0.0304

MAE = 0.0176

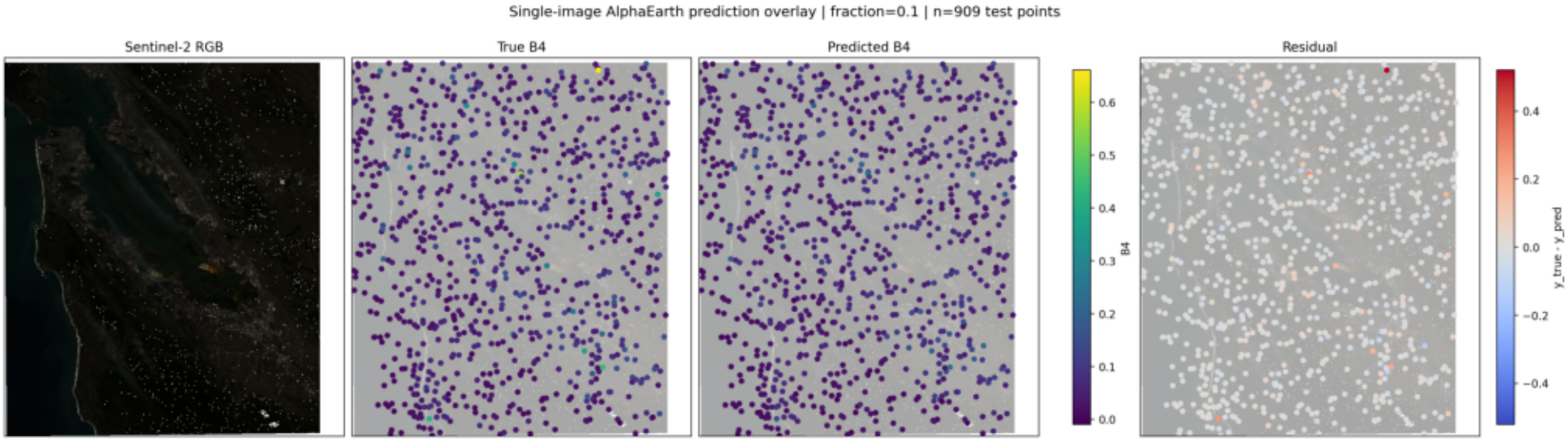
Pearson r = 0.823

Observed pattern:

Performance improves rapidly from 1% to 25% of training rows,

then largely plateaus. On this split, 50% gives the best R^2.

Fraction Sweep Metrics

fraction	train_rows	test_rows	target_col	r2	rmse	mae	pearson_r
0.01	50	909	B4	0.365	0.0426	0.0219	0.619
0.05	182	909	B4	0.578	0.0348	0.0183	0.770
0.10	364	909	B4	0.621	0.0329	0.0170	0.797
0.25	909	909	B4	0.666	0.0309	0.0169	0.817
0.50	1818	909	B4	0.676	0.0304	0.0176	0.823
1.00	3635	909	B4	0.668	0.0308	0.0182	0.821

Held-Out Overlay At Fraction 0.10